Sahaj Bhadja

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Summary

Systems Software Engineer with 2+ years supporting **high-throughput platforms, operating environments, databases, networking, automation, monitoring, and production reliability** through disciplined development, testing, implementation, and incident-response practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Computer Networks, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Computer Architecture, Data Structures, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reworked asynchronous FastAPI orchestration with explicit job states and failure isolation, processing 1K+ concurrent jobs at 3x throughput while sustaining **platform stability, system monitoring, configuration control, and reliable daily operations**.
- Optimized indexed Azure SQL queries and row-level isolation for 500+ users, cutting latency from 1.2 seconds to 350 milliseconds through **database analysis, production debugging, root-cause investigation, and permanent corrective actions**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, processing 1K+ files per batch while reducing turnaround 60% through **automation, failure isolation, testing, and operational troubleshooting**.
- Delivered React and Flask workflows supporting 150+ instructors and 10K+ daily requests while coordinating **application enhancements, implementation procedures, pre-production validation, release monitoring, and technical documentation** across teams.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer for autonomous-driving offline reinforcement learning, improving planning 18% while processing 100K+ trajectories through reproducible **Linux pipelines, high-volume data processing, system benchmarking, and debugging**.
- Architected spatial-temporal encoders combining HD-map geometry, lane context, and ego trajectories, improving closed-loop stability 15% through **memory-aware computation, detailed performance analysis, fault investigation, and documented technical evaluation**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and data pipelines using object-oriented patterns, AWS Glue, S3, and caching, improving response times 30% while reducing technical debt 20% across distributed production components.
- Automated 20+ unit and integration suites across Windows, Linux, and macOS using PyTest, Mockito, Jenkins, and virtual machines, strengthening **change testing, deployment readiness, issue resolution, and operational runbooks**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST services, WebSockets, event-driven processing, and resilient inter-service communication** across distributed locations.
- Automated Docker and Jenkins deployments, reducing release time 50% while using Sentry, Azure Monitor, and Kibana for **system-health monitoring, incident diagnosis, on-call support, and reliable environment maintenance**.

Skills

Mainframe & Systems: IBM Z, z/TPF, TPF Operations Server, HMC, DASD, CEC, LPAR, Linux, Operating Systems
Languages & Automation: Assembler, Java, Python, REXX, OOREXX, Bash, Linux Scripting, C, C++, SQL, Shell Automation
Networking & Integration: TCP/IP, OSA, REST Services, Kafka, JSON, Distributed Systems, Event-Driven Systems, Network Troubleshooting
Operations: Production Support, Monitoring, Configuration Management, Debugging, On-Call Support, Incident Response, Runbooks, ITIL
Engineering: Git, Docker, Jenkins, CI/CD, Unit Testing, Integration Testing, Performance Analysis, Databases, Agile, Technical Documentation

Projects

Mainframe Operations Lab | *Assembler, REXX, Python, z/TPF, TOS, Linux*

- Built a **z/TPF operations simulation environment** that models transaction queues, subsystem health, configuration changes, and recovery procedures while automating monitoring checks, operator commands, runbooks, and incident-response exercises.

Travel Transaction Reliability Engine | *C++, TCP/IP, Kafka, REST APIs, PostgreSQL*

- Designed a **high-throughput travel transaction service** that processes booking events, validates state transitions, isolates failures, replays Kafka messages, monitors latency, and verifies recovery across production-like and pre-production environments.

CPU Scheduling Simulator | *C++, React, JavaScript, Algorithms* | Live

- Implemented performance-sensitive C++ algorithms for FCFS, SJF, SRTF, round-robin, priority, and multilevel queues, validating correctness and visualizing runtime behavior across interactive workloads through a browser-based scheduling simulator.